

Stefano Zanolli

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Education

University of Verona

Verona, Italy

Master's Degree in Computer Engineering (ongoing)

Sep/25 – Jul/27 (expected) · GPA 27/30

- **Relevant Coursework:** Cryptography, Network Security, Malicious Code, Cybersecurity, Software Security.
- **Graduation project (Master's thesis, planned), full-time March – July 2027:** **CRA** (Cyber Resilience Act) and **RED** (Radio Equipment Directive) compliance for embedded devices, with a **focus** on cryptographic mechanisms and security protocols underlying compliance requirements.

University of Verona

Verona, Italy

Bachelor's Degree in Computer Science

Sep/20 – Sep/25

- **Thesis:** First hands-on experience with full-stack web development, optimizing network traffic by splitting the app into microservices behind a reverse proxy.

Work Experience

Edalab

Verona, Italy

Firmware Developer

Mar/26 – Oct/26

- Developed firmware for custom boards from specifications and schematics defined with the client.
- Developed firmware for a CO2 heat-pump water-tank heating system; built a HIL simulator (dual-serial data injection, hydraulic circuit and Modbus inverter/compressor emulation) to test firmware without the real machine.
- Other projects: MQTT/BLE pressure inverter controller, vending machine with spiral dispensers, and CAN firmware updates: adapted a bootloader to a custom board and debugged master-slave (ST-Link, CAN-FD).

VT100 S.r.l.

Verona, Italy

Software Development / IoT Internship

Oct/25 – Jan/26

- Built an **automatic provisioning system** that gives each board its own identity: a MAC from a managed pool and an MQTT identity: credentials or an **X.509 client certificate** from a local **CA**.
- Implemented two enrolment flows (username/password and **mutual TLS**): boards ship with a shared factory identity and swap it on first boot for a unique, revocable one. Stack: Next.js console, EMQX, MySQL, Docker.

Lifeguard

Lazise & Verona, Italy

May/23 – May/24 · Jun/26 – Aug/26

- Responsible for the safety of hundreds of swimmers at high-traffic tourist and sports facilities.

Skills

- **Languages:** Italian (native), English (proficient), German (basic).
- **Firmware Skills:** FreeRTOS, ESP32, STM32, Renesas MCUs (existing codebase), ESP-IDF, communication protocols (USB, CAN, BLE, UART, Modbus, custom protocols), bootloader integration, hardware bring-up (wiring, multimeter/oscilloscope, real-time serial monitoring), debuggers (J-Link/Ozone, ST-Link), soldering (basic level).
- **Firmware Security** (ESP32): Secure Boot, Flash Encryption, Secure OTA, key management, key exchange.
- **Programming Languages:** C, Python, Bash (daily); TypeScript, SQL, HTML/CSS; learning C++, Rust, Assembly.
- **Tools and Frameworks:** Next.js, Prisma, Docker, Git, React.js, Postman, MQTT, Wireshark.

Activities & Interests

Activities: Played the piano for the last 15 years, currently recording a rock album in a studio. I also help organize a few jazz events a year in Verona.

Interests: Gym, swimming and experimenting with new ideas and technologies for personal projects.